

Evaluation Feedback

1. Baseline Synthesis

Incorrect minimum clock period.

The reported minimum clock period is incorrect. The **minimum clock period that satisfies all constraints is 4.5 ns**.

The corresponding PPA metrics are:

- **Area:** 24072 um²
- **Slack:** 0.000 ns
- **Leakage Power:** 4.644e-08 W
- **Dynamic Power:** 0.002859 W

Deduction: -0.5 points

Incorrect VCD back-annotation for power analysis.

The switching activity back-annotation was not performed correctly. The synthesized design uses a **5.5 ns clock period**, while the testbench generates a **2.0 ns clock period**. Since the switching activity is collected during simulation, it must be rescaled using the **simulation clock period**.

In your solution, the `set_power_clock_scaling` command incorrectly specifies the **synthesis clock period** instead of the **simulation clock period**. As a result, the switching activity is not properly scaled, leading to an incorrect dynamic power estimation.

This mistake is present in **both Exercise 1 and Exercise 2**.

Deduction: -0.5 points

2.1 Clock Gating

Incorrect synthesis command for clock gating.

Clock gating should be performed using `compile_ultra`, which enables the appropriate optimization flow for clock-gating insertion. The submitted solution uses `compile` instead, which produces suboptimal results.

Deduction: -0.5 points

2.2 Clock Gating and Multi-VT Synthesis

- **No issues found.**

3. Custom Post-Synthesis Report in PrimeTime

Incorrect use of `get_timing_paths`.

The exercise requires reporting **up to `num_paths` critical paths**, regardless of whether the paths have positive or negative slack.

In the submitted solution, `get_timing_paths` is invoked with the `-nworst` and `-max_paths` options but without specifying `-slack_lesser_than`. In this case, PrimeTime implicitly sets `-slack_lesser_than` to **0**, causing the command to return **only timing paths with negative slack**.

As a result, cells that are traversed exclusively by paths with **positive slack** are not analyzed, making the report incomplete.

To satisfy the exercise requirements, `-slack_lesser_than` should be set to a sufficiently large value (e.g., 1000) so that both positive- and negative-slack paths are considered.

Deduction: -1 points